UNIVERSITI TEKNOLOGI MARA
FINAL EXAMINATION

COURSE	COMPUTER ORGANIZATION AND ARCHITECTURE
COURSE CODE	CSC429
EXAMINATION	NOVEMBER 2026
TIME	3 HOURS

INSTRUCTIONS TO CANDIDATES

1. This question paper consists of two (2) parts: PART A (10 Questions) PART B (1 Question)
2. Answer ALL questions in the Answer Booklet. Start each answer on a new page.
3. Do not bring any material into the examination room unless permission is given by the invigilator.
4. Please check to make sure that this examination pack consists of:
   i) the Question Paper
   ii) an Answer Booklet - provided by the Faculty
5. Answer ALL questions in English.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

This examination paper consists of 6 printed pages.

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PART A (80 MARKS)

Answer ALL questions in this part.

QUESTION A1 **(5 marks)**

Question 1 is based on the following snippet garnered from a computer science web page.

> Henry Edward "Ed" Roberts
> (September 13, 1941 -April 1, 2010)
> was an American engineer, entrepreneur and medical doctor who invented the first commercially successful personal computer in 1975. He is most often known as "the father of the personal computer". The computer was available as a kit for $439 or assembled for $621 and had several additional add-ons such as a memory board and interface boards. By August 1975, over 5,000 Altair 8800 personal computers were sold; starting the personal computer revolution.

a) In which computer generation that a personal computer is introduced?

b) Discuss ONE (1) other major discovery that is found in this generation.

c) Advances in laptop and hand-held devices are accelerating during this computer generation. How such technology is feasible as compared to previous computer generation?

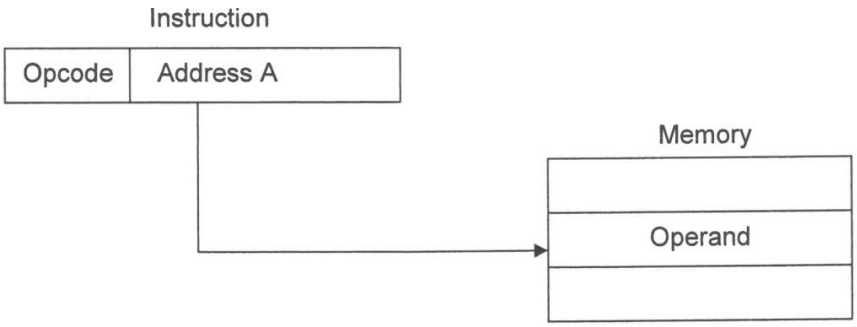
QUESTION A2 **(5 marks)**

Calculate 86 - (-30) using the sign-and-magnitude method. Assume an 8-bit byte. Show your complete workflow.

QUESTION A3 **(5 marks)**

Given the following Figure 2.

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a) Label the addressing technique illustrated in Figure 2.

b) Explain how the addressing technique in 9(a) is used.

List ONE (1) advantage and ONE (1) disadvantage of the addressing technique in

QUESTION A4 **(5 marks)**

Complete the following taxonomy of parallel processor architecture with appropriate example.

Single Instruction Multiple Data Stream ii Processor Organization Multiple Instruction Multiple Data Stream

v

QUESTION A5 **(10 marks)**

Keyboard is the most common means for computer interaction.

a) Explain TWO (2) types of characters that keyboard provides.

b) Describe THREE (3) categories of input and output module. Give an appropriate example for each category.

QUESTION A6 **(10 marks)**

a) Define interrupt.

b) Differentiate between interrupt and direct manipulation access.

c) Given the following scenario. Dilan is surfing woolworth.com to buy groceries online. While he is comparing prices of products in woolworth.com and one other website, two messages popped up onto his screen. Figure 2 (a) and (b) illustrate the conveyed messages.

Are you sure you want to
There is a hardware event failure
in this computer. Please reboot
cancel this request?
this computer now.

(a)

(b)

i. Identify the type of interrupt in Figure 2(a) and Figure 2(b) with proper justification.

ii. While interruptroutinefor Interrupt Ais executing, it is interrupted by interrupt Bwhich has a higher priority than interrupt A. What will happen to interrupt routine A?

QUESTION A7 **(10 marks)**

a) Describe the function of each computer bus type.

b) Discuss TWO (2) benefits of synchronous bus timing.

QUESTION A8 **(10 marks)**

Question 7 is based on the following scenarios. While cleaning his house, Ahmad found an old cassette tape of a popular 80's band. He played the tape and listened to several songs according to the listed tracks. Hafiz was tasked to research on the history of Malay culture by his professor. He went to the National Archive to search for information and manage to access one the resources which was stored in a floppy disk. Sara is planning to buy a new computer. She aims to get one with good processing speed. While researching for computer specifications, she found out that larger cache space has a direct effect towards processing performance. Hana loves playing role-playing games (RPG) and is planning to buy a newly released game that requires a high-speed RAM. But, she is devastated when she found out that her game console does not support the minimum RAM requirement of the game.

a) Based on the above scenarios, identify the access method for each storage devices/memory mentioned in (i), (ii), (iii) and (iv).

b) Describe SRAM and DRAM.

c) Which level of cache is the fastest in a three-level cache architecture? Which type of RAM is used to implement the cache?

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QUESTION A9 **(10 marks)**

a) Discuss the function of each Central Processing Unit (CPU) component'

b) Elaborate on what will happen in the machine cycle if the bus system is faulty

QUESTION A10 **(10 marks)**

Answer both parts. Show your complete workflow.

a) Convert 256610 to hexadecimal.

b) Convert 7E816 to decimal.

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b) Convert 7E816 to decimal.

PART B (20 MARKS)

Answer the question in this part.

QUESTION B1 **(20 marks)**

Question 1 is based on Figure 2 and Table 1.

$$y = A(A + 2B / C - D^2)$$

Write the appropriate programs using correct assembly language notations and comments to compute the equation represented in Figure 1 using zero-, one-, and three-address instructions. The instructions available for use are provided in Table 1. Register A, B, C, and D cannot be modified, and registers x, y, and z can be used as temporary registers. However, the final answer must be stored in Register y.

Zero-address	One-address	Three-address
PUSHp	LOADp	ADD (p,q,r)
POPp	STORp	SUB (p,q,r)
ADD	ADDp	MUL (p.q.r)
SUB	SUBp	DIV(p,q,r)
MUL	MULp	
DIV	DIVp	

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